Noise Effect on the Flux Calibration Parameter $\hat{\ell}$

Symbol Definitions

$$
\begin{cases}
\tilde{\mathbf{b}}_{ij} = \mathbf{b}_{ij} + \boldsymbol{n}_{ij} : \text{ (noisy flux density, } 3\times 1), \qquad \boldsymbol{n}_{ij} = \begin{bmatrix} n_{ij,x} & n_{ij,y} & n_{ij,z} \end{bmatrix}^T \\
\mathbf{g}_j = \Big(\sum_{i=1}^{N_c} \mathbf{S}_i^T \mathbf{S}_i \Big)^{-1} \sum_{i=1}^{N_c} \mathbf{S}_i^T \mathbf{b}_{ij} = \mathbf{W}^{-1} \mathbf{c}_j : \text{ (pole charges, } 6\times 1), \ \mathbf{g}_j = \begin{bmatrix} g_{j1} & \cdots & g_{j6} \end{bmatrix}^T \\
\qquad \text{where: } \ \mathbf{W} = \sum_{i=1}^{N_c} \mathbf{S}_i^T \mathbf{S}_i : \ (6\times 6), \qquad \mathbf{c}_j = \sum_{i=1}^{N_c} \mathbf{S}_i^T \mathbf{b}_{ij} : \ (6\times 1) \\
\tilde{\mathbf{g}}_j = \Big(\sum_{i=1}^{N_c} \mathbf{S}_i^T \mathbf{S}_i \Big)^{-1} \sum_{i=1}^{N_c} \mathbf{S}_i^T \tilde{\mathbf{b}}_{ij} = \mathbf{W}^{-1} \tilde{\mathbf{c}}_j : \text{ (noisy pole charges, } 6\times 1), \ \tilde{\mathbf{g}}_j = \begin{bmatrix} \tilde{g}_{j1} & \cdots & \tilde{g}_{j6} \end{bmatrix}^T \\
\qquad \text{where: } \ \tilde{\mathbf{c}}_j = \sum_{i=1}^{N_c} \mathbf{S}_i^T \tilde{\mathbf{b}}_{ij} : \ (6\times 1) \\
\boldsymbol{\varepsilon}_{ij} = \mathbf{b}_{ij} - \mathbf{S}_i \, \mathbf{g}_j : \text{ (residual, } 3\times 1), \qquad \boldsymbol{\varepsilon}_{ij} = \begin{bmatrix} \varepsilon_{ij,x} & \varepsilon_{ij,y} & \varepsilon_{ij,z} \end{bmatrix}^T \\
\tilde{\boldsymbol{\varepsilon}}_{ij} = \tilde{\mathbf{b}}_{ij} - \mathbf{S}_i \, \tilde{\mathbf{g}}_j : \text{ (noisy residual, } 3\times 1), \qquad \tilde{\boldsymbol{\varepsilon}}_{ij} = \begin{bmatrix} \tilde{\varepsilon}_{ij,x} & \tilde{\varepsilon}_{ij,y} & \tilde{\varepsilon}_{ij,z} \end{bmatrix}^T
\end{cases}
$$

$\rightarrow$.

$$\tilde{J}(\hat{\ell}) = \sum_{j=1}^{N_I} \sum_{i=1}^{N_c} \tilde{\boldsymbol{\varepsilon}}_{ij}^T \tilde{\boldsymbol{\varepsilon}}_{ij}$$

Expanding about $\hat{\ell}_0$ with $\hat{\ell} = \hat{\ell}_0 + \delta\hat{\ell}$:

$$\tilde{J}(\hat{\ell}_0 + \delta\hat{\ell}) = \tilde{J}(\hat{\ell}_0) + \tilde{J}'(\hat{\ell}_0)\, \delta\hat{\ell} + \tfrac{1}{2}\, \tilde{J}''(\hat{\ell}_0)\, \delta\hat{\ell}^{\,2} + O\Big(\delta\hat{\ell}^{\,3}\Big)$$

Since the minimizer of $\tilde{J}$ is $\hat{\ell}_0 + \delta\hat{\ell}$, differentiating with respect to $\delta\hat{\ell}$:

$$\frac{\partial \tilde{J}}{\partial\, \delta\hat{\ell}} = \tilde{J}'(\hat{\ell}_0) + \tilde{J}''(\hat{\ell}_0)\, \delta\hat{\ell} + O\Big(\delta\hat{\ell}^{\,2}\Big) = 0$$

Neglecting the higher-order terms:

$$\delta\hat{\ell} \approx -\frac{\tilde{J}'(\hat{\ell}_0)}{\tilde{J}''(\hat{\ell}_0)}$$

$\rightarrow$. Numerator: First Derivative $\tilde{J}'(\hat{\ell}_0)$

A prime denotes $d/d\hat{\ell}$.

$$\tilde{J}' = 2 \sum_{j=1}^{N_I} \sum_{i=1}^{N_c} \tilde{\boldsymbol{\varepsilon}}_{ij}^T \tilde{\boldsymbol{\varepsilon}}_{ij}'$$

Since $\tilde{\mathbf{b}}_{ij}$ does not depend on $\hat{\ell}$:

$$\tilde{\boldsymbol{\varepsilon}}_{ij}' = -\mathbf{S}_i' \, \tilde{\mathbf{g}}_j - \mathbf{S}_i \, \tilde{\mathbf{g}}_j'$$

Differentiating $\mathbf{W}\tilde{\mathbf{g}}_j = \tilde{\mathbf{c}}_j$:

$$\tilde{\mathbf{g}}'_j = \mathbf{W}^{-1}\left(\tilde{\mathbf{c}}'_j - \mathbf{W}'\tilde{\mathbf{g}}_j\right) = \mathbf{W}^{-1}\sum_{i=1}^{N_c}\left(\mathbf{S}_i'^T\tilde{\boldsymbol{\varepsilon}}_{ij} - \mathbf{S}_i^T\mathbf{S}'_i\,\tilde{\mathbf{g}}_j\right)$$

where:

$$\mathbf{W}' = \sum_{i=1}^{N_c}\left(\mathbf{S}_i'^T\mathbf{S}_i + \mathbf{S}_i^T\mathbf{S}'_i\right), \qquad \tilde{\mathbf{c}}'_j = \sum_{i=1}^{N_c}\mathbf{S}_i'^T\tilde{\mathbf{b}}_{ij}$$

Substituting $\tilde{\mathbf{g}}'_j$ into $\tilde{\boldsymbol{\varepsilon}}'_{ij}$:

$$\tilde{\boldsymbol{\varepsilon}}'_{ij} = -\mathbf{S}'_i\,\tilde{\mathbf{g}}_j - \mathbf{S}_i\mathbf{W}^{-1}\left(\tilde{\mathbf{c}}'_j - \mathbf{W}'\tilde{\mathbf{g}}_j\right)$$

Substituting $\tilde{\boldsymbol{\varepsilon}}'_{ij}$ into $\tilde{J}'$:

$$\tilde{J}' = -2\sum_{j=1}^{N_I}\sum_{i=1}^{N_c}\tilde{\boldsymbol{\varepsilon}}_{ij}^T\left[\mathbf{S}'_i\,\tilde{\mathbf{g}}_j + \mathbf{S}_i\mathbf{W}^{-1}\left(\tilde{\mathbf{c}}'_j - \mathbf{W}'\tilde{\mathbf{g}}_j\right)\right]$$

Expanding:

$$\begin{aligned}\tilde{J}' = &-2\sum_{j=1}^{N_I}\sum_{i=1}^{N_c}\tilde{\boldsymbol{\varepsilon}}_{ij}^T\mathbf{S}'_i\,\tilde{\mathbf{g}}_j - 2\sum_{j=1}^{N_I}\sum_{i=1}^{N_c}\tilde{\boldsymbol{\varepsilon}}_{ij}^T\mathbf{S}_i\mathbf{W}^{-1}\tilde{\mathbf{c}}'_j \\ &+ 2\sum_{j=1}^{N_I}\sum_{i=1}^{N_c}\tilde{\boldsymbol{\varepsilon}}_{ij}^T\mathbf{S}_i\mathbf{W}^{-1}\mathbf{W}'\tilde{\mathbf{g}}_j\end{aligned}$$

According to $\mathbf{W}\tilde{\mathbf{g}}_j = \tilde{\mathbf{c}}_j$:

$$\sum_{i=1}^{N_c}\tilde{\boldsymbol{\varepsilon}}_{ij}^T\mathbf{S}_i = \Big(\sum_{i=1}^{N_c}\mathbf{S}_i^T\tilde{\boldsymbol{\varepsilon}}_{ij}\Big)^T = \left(\tilde{\mathbf{c}}_j - \mathbf{W}\tilde{\mathbf{g}}_j\right)^T = \mathbf{0}^T$$

so the last two terms vanish:

$$\tilde{J}' = -2\sum_{j=1}^{N_I}\sum_{i=1}^{N_c}\tilde{\boldsymbol{\varepsilon}}_{ij}^T\mathbf{S}'_i\,\tilde{\mathbf{g}}_j$$

三. Noise Propagation into $\tilde{\mathbf{g}}_j$ and $\tilde{\boldsymbol{\varepsilon}}_{ij}$

Substituting $\tilde{\mathbf{b}}_{ij} = \mathbf{b}_{ij} + n_{ij}$ into $\tilde{\mathbf{g}}_j$:

$$\tilde{\mathbf{g}}_j = \mathbf{W}^{-1}\sum_{i=1}^{N_c}\mathbf{S}_i^T\left(\mathbf{b}_{ij} + n_{ij}\right) = \underbrace{\mathbf{W}^{-1}\sum_{i=1}^{N_c}\mathbf{S}_i^T\mathbf{b}_{ij}}_{\mathbf{g}_j\ (6\times1)} + \underbrace{\mathbf{W}^{-1}\sum_{i=1}^{N_c}\mathbf{S}_i^T n_{ij}}_{\delta\mathbf{g}_j\ (6\times1)} = \mathbf{g}_j + \delta\mathbf{g}_j$$

Substituting $\tilde{\mathbf{b}}_{ij} = \mathbf{b}_{ij} + n_{ij}$ and $\tilde{\mathbf{g}}_j = \mathbf{g}_j + \delta\mathbf{g}_j$ into $\tilde{\boldsymbol{\varepsilon}}_{ij}$:

$$\tilde{\boldsymbol{\varepsilon}}_{ij} = \left(\mathbf{b}_{ij} + n_{ij}\right) - \mathbf{S}_i\left(\mathbf{g}_j + \delta\mathbf{g}_j\right) = \underbrace{\mathbf{b}_{ij} - \mathbf{S}_i\,\mathbf{g}_j}_{\boldsymbol{\varepsilon}_{ij}\ (3\times1)} + \underbrace{n_{ij} - \mathbf{S}_i\,\delta\mathbf{g}_j}_{\delta\boldsymbol{\varepsilon}_{ij}\ (3\times1)} = \boldsymbol{\varepsilon}_{ij} + \delta\boldsymbol{\varepsilon}_{ij}$$

Substituting $\tilde{\mathbf{g}}_j$ and $\tilde{\boldsymbol{\varepsilon}}_{ij}$ into $\tilde{J}'$, evaluated at $\hat{\ell}_0$:

$$\begin{aligned}\tilde{J}'(\hat{\ell}_0) &= -2\sum_{j=1}^{N_I}\sum_{i=1}^{N_c}\left(\boldsymbol{\varepsilon}_{ij}+\delta\boldsymbol{\varepsilon}_{ij}\right)^T\mathbf{S}_i'\left(\mathbf{g}_j+\delta\mathbf{g}_j\right)\\ &= \sum_{j=1}^{N_I}\Bigg[\underbrace{-2\sum_{i=1}^{N_c}\boldsymbol{\varepsilon}_{ij}^T\mathbf{S}_i'\,\mathbf{g}_j}_{H_{0,j}}\ \underbrace{-2\sum_{i=1}^{N_c}\left(\delta\boldsymbol{\varepsilon}_{ij}^T\mathbf{S}_i'\,\mathbf{g}_j+\boldsymbol{\varepsilon}_{ij}^T\mathbf{S}_i'\,\delta\mathbf{g}_j\right)}_{H_{1,j}}\ \underbrace{-2\sum_{i=1}^{N_c}\delta\boldsymbol{\varepsilon}_{ij}^T\mathbf{S}_i'\,\delta\mathbf{g}_j}_{H_{2,j}}\Bigg]\\ &= \sum_{j=1}^{N_I}\left(H_{0,j}+H_{1,j}+H_{2,j}\right)\end{aligned}$$

where $H_{k,j}$ $(k=0,1,2)$: (k-th order term in the noise of the j-th excitation, scalar).
$H_{0,j}$ contains no noise; it is $\tilde{J}'$ with $\tilde{\mathbf{b}}_{ij}$ replaced by $\mathbf{b}_{ij}$. Since $\hat{\ell}_0$ is the minimizer of the noise-free cost,

$$\sum_{j=1}^{N_I}H_{0,j}=J'(\hat{\ell}_0)=0$$

so

$$\tilde{J}'(\hat{\ell}_0)=\sum_{j=1}^{N_I}\left(H_{1,j}+H_{2,j}\right)$$

四. Denominator: Second Derivative $\tilde{J}''(\hat{\ell}_0)$

Differentiating $\tilde{J}'=2\sum_j\sum_i\tilde{\boldsymbol{\varepsilon}}_{ij}^T\tilde{\boldsymbol{\varepsilon}}_{ij}'$:

$$\tilde{J}''=2\sum_{j=1}^{N_I}\sum_{i=1}^{N_c}\left(\tilde{\boldsymbol{\varepsilon}}_{ij}'^T\tilde{\boldsymbol{\varepsilon}}_{ij}'+\tilde{\boldsymbol{\varepsilon}}_{ij}^T\tilde{\boldsymbol{\varepsilon}}_{ij}''\right)$$

Differentiating $\tilde{\boldsymbol{\varepsilon}}_{ij}'=-\mathbf{S}_i'\,\tilde{\mathbf{g}}_j-\mathbf{S}_i\,\tilde{\mathbf{g}}_j'$:

$$\tilde{\boldsymbol{\varepsilon}}_{ij}''=-\left(\mathbf{S}_i''\,\tilde{\mathbf{g}}_j+\mathbf{S}_i'\,\tilde{\mathbf{g}}_j'\right)-\left(\mathbf{S}_i'\,\tilde{\mathbf{g}}_j'+\mathbf{S}_i\,\tilde{\mathbf{g}}_j''\right)=-\mathbf{S}_i''\,\tilde{\mathbf{g}}_j-2\,\mathbf{S}_i'\,\tilde{\mathbf{g}}_j'-\mathbf{S}_i\,\tilde{\mathbf{g}}_j''$$

Substituting $\tilde{\boldsymbol{\varepsilon}}_{ij}''$ into $\tilde{J}''$ and expanding:

$$\begin{aligned}\tilde{J}''=2\sum_{j=1}^{N_I}\sum_{i=1}^{N_c}\|\tilde{\boldsymbol{\varepsilon}}_{ij}'\|^2-2\sum_{j=1}^{N_I}\sum_{i=1}^{N_c}\tilde{\boldsymbol{\varepsilon}}_{ij}^T\mathbf{S}_i''\,\tilde{\mathbf{g}}_j\\ -4\sum_{j=1}^{N_I}\sum_{i=1}^{N_c}\tilde{\boldsymbol{\varepsilon}}_{ij}^T\mathbf{S}_i'\,\tilde{\mathbf{g}}_j'-2\sum_{j=1}^{N_I}\sum_{i=1}^{N_c}\tilde{\boldsymbol{\varepsilon}}_{ij}^T\mathbf{S}_i\,\tilde{\mathbf{g}}_j''\end{aligned}$$

According to $\mathbf{W}\tilde{\mathbf{g}}_j=\tilde{\mathbf{c}}_j$, $\sum_i\tilde{\boldsymbol{\varepsilon}}_{ij}^T\mathbf{S}_i=\mathbf{0}^T$, so the last term vanishes:

$$\tilde{J}''=2\sum_{j=1}^{N_I}\sum_{i=1}^{N_c}\|\tilde{\boldsymbol{\varepsilon}}_{ij}'\|^2-2\sum_{j=1}^{N_I}\sum_{i=1}^{N_c}\tilde{\boldsymbol{\varepsilon}}_{ij}^T\mathbf{S}_i''\,\tilde{\mathbf{g}}_j-4\sum_{j=1}^{N_I}\sum_{i=1}^{N_c}\tilde{\boldsymbol{\varepsilon}}_{ij}^T\mathbf{S}_i'\,\tilde{\mathbf{g}}_j'$$

Substituting $\tilde{\mathbf{b}}_{ij} = \mathbf{b}_{ij} + n_{ij}$ and $\tilde{\mathbf{g}}_j = \mathbf{g}_j + \delta\mathbf{g}_j$ into $\tilde{\mathbf{g}}'_j$:

$$\tilde{\mathbf{g}}'_j = \mathbf{W}^{-1}\left(\tilde{\mathbf{c}}'_j - \mathbf{W}'\tilde{\mathbf{g}}_j\right) = \mathbf{W}^{-1}\Big[\sum_{i=1}^{N_c}\mathbf{S}_i'^T\left(\mathbf{b}_{ij} + n_{ij}\right) - \mathbf{W}'\left(\mathbf{g}_j + \delta\mathbf{g}_j\right)\Big]$$
$$= \underbrace{\mathbf{W}^{-1}\Big(\sum_{i=1}^{N_c}\mathbf{S}_i'^T\mathbf{b}_{ij} - \mathbf{W}'\mathbf{g}_j\Big)}_{\mathbf{g}'_j\ (6\times1)} + \underbrace{\mathbf{W}^{-1}\Big(\sum_{i=1}^{N_c}\mathbf{S}_i'^T n_{ij} - \mathbf{W}'\delta\mathbf{g}_j\Big)}_{\delta\mathbf{g}'_j\ (6\times1)} = \mathbf{g}'_j + \delta\mathbf{g}'_j$$

Substituting $\delta\mathbf{g}_j = \mathbf{W}^{-1}\sum_i \mathbf{S}_i^T n_{ij}$ into $\delta\mathbf{g}'_j$:

$$\delta\mathbf{g}'_j = \mathbf{W}^{-1}\Big(\sum_{i=1}^{N_c}\mathbf{S}_i'^T n_{ij} - \mathbf{W}'\mathbf{W}^{-1}\sum_{i=1}^{N_c}\mathbf{S}_i^T n_{ij}\Big) = \mathbf{W}^{-1}\sum_{i=1}^{N_c}\left(\mathbf{S}_i'^T - \mathbf{W}'\mathbf{W}^{-1}\mathbf{S}_i^T\right) n_{ij}$$

Substituting $\tilde{\mathbf{g}}_j = \mathbf{g}_j + \delta\mathbf{g}_j$ and $\tilde{\mathbf{g}}'_j = \mathbf{g}'_j + \delta\mathbf{g}'_j$ into $\tilde{\boldsymbol{\varepsilon}}'_{ij}$:

$$\tilde{\boldsymbol{\varepsilon}}'_{ij} = -\mathbf{S}'_i\left(\mathbf{g}_j + \delta\mathbf{g}_j\right) - \mathbf{S}_i\left(\mathbf{g}'_j + \delta\mathbf{g}'_j\right) = \underbrace{-\mathbf{S}'_i\,\mathbf{g}_j - \mathbf{S}_i\,\mathbf{g}'_j}_{\boldsymbol{\varepsilon}'_{ij}\ (3\times1)}\ \underbrace{-\,\mathbf{S}'_i\,\delta\mathbf{g}_j - \mathbf{S}_i\,\delta\mathbf{g}'_j}_{\delta\boldsymbol{\varepsilon}'_{ij}\ (3\times1)} = \boldsymbol{\varepsilon}'_{ij} + \delta\boldsymbol{\varepsilon}'_{ij}$$

Substituting all decompositions into $\tilde{J}''$, evaluated at $\hat{\ell}_0$:

$$\begin{aligned}
\tilde{J}''(\hat{\ell}_0) &= 2\sum_{j=1}^{N_I}\sum_{i=1}^{N_c}\|\boldsymbol{\varepsilon}'_{ij} + \delta\boldsymbol{\varepsilon}'_{ij}\|^2 - 2\sum_{j=1}^{N_I}\sum_{i=1}^{N_c}\left(\boldsymbol{\varepsilon}_{ij} + \delta\boldsymbol{\varepsilon}_{ij}\right)^T\mathbf{S}''_i\left(\mathbf{g}_j + \delta\mathbf{g}_j\right) \\
&\quad - 4\sum_{j=1}^{N_I}\sum_{i=1}^{N_c}\left(\boldsymbol{\varepsilon}_{ij} + \delta\boldsymbol{\varepsilon}_{ij}\right)^T\mathbf{S}'_i\left(\mathbf{g}'_j + \delta\mathbf{g}'_j\right) \\
&= \sum_{j=1}^{N_I}\Bigg[\underbrace{2\sum_{i=1}^{N_c}\|\boldsymbol{\varepsilon}'_{ij}\|^2 - 2\sum_{i=1}^{N_c}\boldsymbol{\varepsilon}_{ij}^T\mathbf{S}''_i\,\mathbf{g}_j - 4\sum_{i=1}^{N_c}\boldsymbol{\varepsilon}_{ij}^T\mathbf{S}'_i\,\mathbf{g}'_j}_{D_{0,j}} \\
&\qquad \underbrace{+\,4\sum_{i=1}^{N_c}\boldsymbol{\varepsilon}_{ij}'^T\delta\boldsymbol{\varepsilon}'_{ij} - 2\sum_{i=1}^{N_c}\left(\delta\boldsymbol{\varepsilon}_{ij}^T\mathbf{S}''_i\,\mathbf{g}_j + \boldsymbol{\varepsilon}_{ij}^T\mathbf{S}''_i\,\delta\mathbf{g}_j\right) - 4\sum_{i=1}^{N_c}\left(\delta\boldsymbol{\varepsilon}_{ij}^T\mathbf{S}'_i\,\mathbf{g}'_j + \boldsymbol{\varepsilon}_{ij}^T\mathbf{S}'_i\,\delta\mathbf{g}'_j\right)}_{D_{1,j}} \\
&\qquad \underbrace{+\,2\sum_{i=1}^{N_c}\|\delta\boldsymbol{\varepsilon}'_{ij}\|^2 - 2\sum_{i=1}^{N_c}\delta\boldsymbol{\varepsilon}_{ij}^T\mathbf{S}''_i\,\delta\mathbf{g}_j - 4\sum_{i=1}^{N_c}\delta\boldsymbol{\varepsilon}_{ij}^T\mathbf{S}'_i\,\delta\mathbf{g}'_j}_{D_{2,j}}\Bigg] \\
&= \sum_{j=1}^{N_I}\left(D_{0,j} + D_{1,j} + D_{2,j}\right)
\end{aligned}$$

where $D_{k,j}$ $(k = 0, 1, 2)$: (k-th order term in the noise of the j-th excitation, scalar).

II. Linearization in the Noise

Substituting $\tilde{J}'(\hat{\ell}_0)$ and $\tilde{J}''(\hat{\ell}_0)$ into $\delta\hat{\ell}$:

$$\delta\hat{\ell} \approx -\frac{\tilde{J}'(\hat{\ell}_0)}{\tilde{J}''(\hat{\ell}_0)} = -\frac{\sum_j\left(H_{1,j} + H_{2,j}\right)}{\sum_j\left(D_{0,j} + D_{1,j} + D_{2,j}\right)}$$

Let $H_k = \sum_j H_{k,j}$ and $D_k = \sum_j D_{k,j}$ $(k = 0, 1, 2)$:

$$\delta\hat{\ell} \approx -\frac{H_1 + H_2}{D_0 + D_1 + D_2} = -\frac{H_1 + H_2}{D_0} \cdot \frac{1}{1+x}, \qquad x = \frac{D_1 + D_2}{D_0}$$

Since x is small, $\frac{1}{1+x} = 1 - x + x^2 - \cdots$, and grouping by order in the noise:

$$\delta\hat{\ell} = \underbrace{-\frac{H_1}{D_0}}_{O(n)} + \underbrace{\left(-\frac{H_2}{D_0} + \frac{H_1 D_1}{D_0^2}\right)}_{O(n^2)} + O\big(n^3\big)$$

Keeping only the first-order term:

$$\delta\hat{\ell} \approx -\frac{H_1}{D_0} = -\frac{\sum_j H_{1,j}}{\sum_j D_{0,j}}$$

From $H_{1,j}$:

$$H_1 = \sum_{j=1}^{N_I} H_{1,j} = -2\sum_{j=1}^{N_I}\sum_{i=1}^{N_c} \left(\delta\boldsymbol{\varepsilon}_{ij}^T \mathbf{S}_i' \,\mathbf{g}_j + \boldsymbol{\varepsilon}_{ij}^T \mathbf{S}_i' \,\delta\mathbf{g}_j\right)$$

Substituting $\delta\boldsymbol{\varepsilon}_{ij} = \boldsymbol{n}_{ij} - \mathbf{S}_i \,\delta\mathbf{g}_j$ (and $\boldsymbol{\varepsilon}_{ij}^T \mathbf{S}_i' \,\delta\mathbf{g}_j = \delta\mathbf{g}_j^T \mathbf{S}_i'^T \boldsymbol{\varepsilon}_{ij}$, a scalar):

$$\begin{aligned} H_1 &= -2\sum_{j=1}^{N_I}\sum_{i=1}^{N_c} \left(\boldsymbol{n}_{ij}^T \mathbf{S}_i' \,\mathbf{g}_j - \delta\mathbf{g}_j^T \mathbf{S}_i^T \mathbf{S}_i' \,\mathbf{g}_j + \delta\mathbf{g}_j^T \mathbf{S}_i'^T \boldsymbol{\varepsilon}_{ij}\right) \\ &= -2\sum_{j=1}^{N_I} \Big[\sum_{i=1}^{N_c} \boldsymbol{n}_{ij}^T \mathbf{S}_i' \,\mathbf{g}_j + \delta\mathbf{g}_j^T \underbrace{\sum_{i=1}^{N_c} \left(\mathbf{S}_i'^T \boldsymbol{\varepsilon}_{ij} - \mathbf{S}_i^T \mathbf{S}_i' \,\mathbf{g}_j\right)}_{\mathbf{W}\mathbf{g}_j'} \Big] \end{aligned}$$

where $\mathbf{g}_j' = \mathbf{W}^{-1} \sum_i \left(\mathbf{S}_i'^T \boldsymbol{\varepsilon}_{ij} - \mathbf{S}_i^T \mathbf{S}_i' \,\mathbf{g}_j\right)$ has the same form as $\tilde{\mathbf{g}}_j'$.
Substituting $\delta\mathbf{g}_j = \mathbf{W}^{-1} \sum_i \mathbf{S}_i^T \boldsymbol{n}_{ij}$ ($\mathbf{W}$ symmetric):

$$\delta\mathbf{g}_j^T \mathbf{W} \mathbf{g}_j' = \Big(\sum_{i=1}^{N_c} \boldsymbol{n}_{ij}^T \mathbf{S}_i\Big) \mathbf{W}^{-1} \mathbf{W} \mathbf{g}_j' = \sum_{i=1}^{N_c} \boldsymbol{n}_{ij}^T \mathbf{S}_i \,\mathbf{g}_j'$$

so, with $\boldsymbol{\varepsilon}_{ij}' = -\mathbf{S}_i' \,\mathbf{g}_j - \mathbf{S}_i \,\mathbf{g}_j'$:

$$H_1 = -2\sum_{j=1}^{N_I}\sum_{i=1}^{N_c} \boldsymbol{n}_{ij}^T \left(\mathbf{S}_i' \,\mathbf{g}_j + \mathbf{S}_i \,\mathbf{g}_j'\right) = 2\sum_{j=1}^{N_I}\sum_{i=1}^{N_c} \boldsymbol{\varepsilon}_{ij}'^T \boldsymbol{n}_{ij}$$

From $D_{0,j}$:

$$D_0 = \sum_{j=1}^{N_I} D_{0,j} = 2\sum_{j=1}^{N_I}\sum_{i=1}^{N_c} \|\boldsymbol{\varepsilon}_{ij}'\|^2 - 2\sum_{j=1}^{N_I}\sum_{i=1}^{N_c} \boldsymbol{\varepsilon}_{ij}^T \mathbf{S}_i'' \,\mathbf{g}_j - 4\sum_{j=1}^{N_I}\sum_{i=1}^{N_c} \boldsymbol{\varepsilon}_{ij}^T \mathbf{S}_i' \,\mathbf{g}_j'$$

This is $\tilde{J}''$ with $\tilde{\mathbf{b}}_{ij}$ replaced by $\mathbf{b}_{ij}$, i.e. the second derivative of the noise-free cost:

$$D_0 = J''(\hat{\ell}_0)$$

Substituting H_1 and D_0 into $\delta\hat{\ell} \approx -H_1/D_0$:

$$\delta\hat{\ell} \approx -\frac{2}{J''(\hat{\ell}_0)} \sum_{j=1}^{N_I}\sum_{i=1}^{N_c} \boldsymbol{\varepsilon}_{ij}'^T \boldsymbol{n}_{ij} = \boldsymbol{\mathcal{E}}^T \mathbf{N}$$

where:

$$\boldsymbol{\mathcal{E}} = -\frac{2}{J''(\hat{\ell}_0)} \begin{bmatrix} \boldsymbol{\varepsilon}'_{11} \\ \boldsymbol{\varepsilon}'_{21} \\ \vdots \\ \boldsymbol{\varepsilon}'_{N_c N_I} \end{bmatrix} : \ (3N_c N_I \times 1), \qquad \mathbf{N} = \begin{bmatrix} \boldsymbol{n}_{11} \\ \boldsymbol{n}_{21} \\ \vdots \\ \boldsymbol{n}_{N_c N_I} \end{bmatrix} : \ (3N_c N_I \times 1)$$

六. Standard Deviation of $\hat{\ell}$

Noise assumptions:

$$E\left[\boldsymbol{n}_{ij}\right] = \mathbf{0}, \qquad \mathrm{Cov}\left[\boldsymbol{n}_{ij}\right] = \sigma^2 \mathbf{E}_3$$

and $\boldsymbol{n}_{ij}$ are independent for different i and j, where σ is the noise standard deviation of each component and $\mathbf{E}_3$ is the 3×3 identity matrix.
Taking the variance of $\delta\hat{\ell}$:

$$\sigma^2_{\delta\hat{\ell}} = \sigma^2_{\hat{\ell}} = \mathrm{Var}\Big(-\frac{2}{J''(\hat{\ell}_0)} \sum_{j=1}^{N_I} \sum_{i=1}^{N_c} \boldsymbol{\varepsilon}'^{T}_{ij} \boldsymbol{n}_{ij} \Big)$$

Since $\mathrm{Var}(cX) = c^2 \,\mathrm{Var}(X)$ for a constant c, the coefficient comes out squared:

$$\sigma^2_{\hat{\ell}} = \frac{4}{J''(\hat{\ell}_0)^2} \mathrm{Var}\Big(\sum_{j=1}^{N_I} \sum_{i=1}^{N_c} \boldsymbol{\varepsilon}'^{T}_{ij} \boldsymbol{n}_{ij} \Big)$$

Since the $\boldsymbol{n}_{ij}$ are independent, the variance of the sum is the sum of the variances:

$$\sigma^2_{\hat{\ell}} = \frac{4}{J''(\hat{\ell}_0)^2} \sum_{j=1}^{N_I} \sum_{i=1}^{N_c} \mathrm{Var}\left(\boldsymbol{\varepsilon}'^{T}_{ij} \boldsymbol{n}_{ij} \right)$$

Since $\mathrm{Var}\left(\boldsymbol{\varepsilon}'^{T}_{ij} \boldsymbol{n}_{ij} \right) = \boldsymbol{\varepsilon}'^{T}_{ij} \,\mathrm{Cov}\left[\boldsymbol{n}_{ij}\right] \boldsymbol{\varepsilon}'_{ij} = \sigma^2 \|\boldsymbol{\varepsilon}'_{ij}\|^2$:

$$\sigma^2_{\hat{\ell}} = \frac{4\sigma^2}{J''(\hat{\ell}_0)^2} \sum_{j=1}^{N_I} \sum_{i=1}^{N_c} \|\boldsymbol{\varepsilon}'_{ij}\|^2$$

Taking the square root ($J''(\hat{\ell}_0) > 0$ at the minimizer):

$$\sigma_{\hat{\ell}} = \sigma_{\delta\hat{\ell}} = \underbrace{\frac{2\sqrt{\sum_{j=1}^{N_I} \sum_{i=1}^{N_c} \|\boldsymbol{\varepsilon}'_{ij}\|^2}}{J''(\hat{\ell}_0)}}_{\eta_{\hat{\ell}}} \sigma = \eta_{\hat{\ell}} \, \sigma$$

where $\eta_{\hat{\ell}}$: (sensitivity of $\hat{\ell}$ to the sensor noise, μm/mT).